

NANDINI SAXENA

Curriculum Vitae

PERSONAL INFORMATION

Address: Pune, Maharashtra, India	Email: nandinisaxenawork@gmail.com
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Medium: medium.com/@nandinisaxenawork	Nationality: Indian

ABOUT ME

Final-year Computer Science bachelor's student (CGPA 9.3/10) focused on machine learning and data science, with hands-on research experience in navigation under sensor failure, synthetic data integrity, fraud detection under concept drift, and applied computer vision. Seeking a **Pflichtpraktikum** (approximately 16-18 weeks) in Berlin, **January to April 2027**, in any **data science, machine learning, or applied AI setting**, not tied to a single domain. Combines independent technical project work with team leadership experience running a 1500+ member developer community.

WORK EXPERIENCE

Machine Learning Intern — DIAT, Defence Research and Development Organisation (DRDO), Pune *Jan 2026 – Apr 2026*

- Built a transformer-based sequence model for UAV trajectory reconstruction during GPS outages using IMU sensor data, reducing trajectory RMSE by ~28% versus classical dead-reckoning integration
- Designed the dead-reckoning pipeline end-to-end, noise filtering, bias correction, and drift modelling for inertial-only navigation under GPS denial

Vice President — Google Developer Groups (GDG) BVP Pune *Jun 2025 – Sept 2026*

- Lead a 12-person core team running technical programming for a 1500+ member developer community
- Secured and manage 5+ industry partnerships for events, sponsorships, and speaker engagements

Software Development Engineer Intern — MCarbon Tech *Jun 2025 – Jul 2025*

- Built a smart ticketing system on Spring Boot, Flask, and Vue.js, covering backend services, API integration, and frontend delivery

EDUCATION AND TRAINING

BTech, Computer Science & Engineering — **CGPA 9.3/10** — Bharati Vidyapeeth (Deemed to be University), College of Engineering, Pune *Jul 2023 – June 2027*

Relevant coursework: Machine Learning, Deep Learning, Natural Language Processing, Agentic AI, Business Intelligence, Cloud Computing

One-year gap between higher secondary education and the start of this degree (2022–2023), spent preparing for undergraduate engineering entrance examinations.

Higher Secondary Certificate (Class XII) — **82.6%** — Delhi Public School, Hisar *April 2021 – March 2022*

Secondary School Certificate (Class X) — **96%** — Delhi Public School, Hisar *April 2019 – March 2020*

PERSONAL SKILLS

Mother tongue: Hindi

Other languages

Language	Listening	Reading	Spoken interaction	Spoken production	Writing
English	C1	C1	C1	C1	C1
German	B1	B1	B1	B1	B1

Self-assessed against the Common European Framework of Reference for Languages (CEFR).

Communication skills: Strong written and verbal communication, developed through technical writing (published explainers on Medium) and leading partner/sponsor conversations as GDG Vice President.

Organisational / managerial skills: Lead a 12-person core team and coordinate programming for a 1500+ member community; manage 5+ concurrent industry partnerships end-to-end (outreach, negotiation, delivery).

Job-related skills

- Languages: Python, Java, SQL, JavaScript, C
- ML / DL: PyTorch, LightGBM, XGBoost, scikit-learn, HuggingFace Transformers
- Focus areas: time-series forecasting, sequence modeling (LSTM, TCN+attention), model interpretability (SHAP), experiment design (temporal cross-validation, ablation studies, statistical significance testing)
- Backend / Infra: FastAPI, Flask, Spring Boot, PostgreSQL, Docker, Apache Airflow, Terraform, AWS

Digital skills

Information & data literacy	Communication & collaboration	Content creation	Safety	Problem solving
Proficient	Proficient	Proficient	Independent	Proficient

Self-assessed against the European Digital Competence Framework (DigComp).

ADDITIONAL INFORMATION

Selected projects (github.com/nandini1612)

1. [GateIO](#)

- a. Built a yaw-rate-gated deep learning system (PyTorch; LSTM, TCN+attention) predicting UAV velocity through GPS outages, cutting mean position drift to 11.2 m over 10 s outages — 7.5x better than a tuned Extended Kalman Filter (MARS-LVIG dataset)
- b. Diagnosed a data-leakage flaw in standard within-flight evaluation splits (~3x inflated accuracy) and introduced a leakage-free leave-one-flight-out cross-validation protocol

2. [Fraud Detection](#)

- a. Deployed an end-to-end fraud detection service (LightGBM, FastAPI) on 590K IEEE-CIS transactions with calibrated probabilities, per-prediction SHAP explanations, and a 32-test suite; Dockerized with Prometheus metrics and PSI-based drift monitoring
- b. Ran a temporally-split ablation across 4 model families and 3 fraud datasets, showing velocity features top SHAP importance yet add no significant predictive value once graph features are present (Wilcoxon $p = 0.23, 0.95$) — while graph features are the real driver (AUPRC +0.031, $p = 0.002$)

3. [CashPulse](#)

- a. Built a cash-flow analytics pipeline with parallel ETL (SQLite) and ELT (Postgres + dbt) tiers, orchestrated with Airflow and provisioned with Terraform
- b. Forecasts cash flow and flags anomalies (gradient-boosted regressor, Isolation Forest), gated by 54 automated tests across 2 CI pipelines; deployed as an authenticated Streamlit dashboard

4. [Phenological Mismatch Observatory](#)

- a. validated a weakly-supervised phenology classifier against five years of independent ground-truth records rather than a held-out split of the same dataset (6.7-day MAE)

Working Papers

- [GateIO](#): Yaw-rated-gated Casual Deep Learning for UAV GPS-Outage Bridging - preprint on Zenodo (DOI 10.5281/zenodo.22853978); preparing submission to IEEE/ION PLANS 2027
- **Importance Is Not Contribution**: Velocity Features Are Redundant on Real Fraud Data but Rewarded by Simulated Benchmarks – preprint on Zenodo (DOI 10.5281/zenodo.22892826)
- Disentangling Logical Validity from Statistical Utility in Synthetic Tabular Data - manuscript in preparation

Writing (Medium, [@nandinisaxenawork](#))

- [Explainers on Prophet's seasonality decomposition](#), [GNNs vs. simpler models under distribution drift](#), [CNN inductive biases](#) and [Markov chains](#)

Leadership and activities

- Founder, OpporYouNity — youth opportunity & mental-health platform
- Head of PR, Null Student Chapter
- Classical Bharatanatyam dancer, 10+ years (RAQS College Dance Team)